Pricing, Hedging, and Model Risk

Vanilla claims, path dependence, and stochastic discounting

Lecture 13

A price is only half of a model

The question

A model assigns a price, but it also prescribes a hedge. How should we judge the hedge when the market evolves differently from the model?

1. Start with a vanilla call and the Black–Scholes replication benchmark.
2. Change the model while keeping the contract fixed.
3. Change the contract while keeping the market model simple.
4. Add stochastic rates and ask which traded assets span the risks.

NO-ARBITRAGE A pricing model should be assessed both by the prices it fits and by the hedging P&L it generates outside the model.

A vanilla option: the hedge depends on the model

For a European call, Black–Scholes links price and replication

The payoff is

$$(S_T - K)^+.$$

Under the risk-neutral Black–Scholes model,

$$\frac{\mathrm{d}S_t}{S_t} = r\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^{\mathbb{Q}},$$

the option price is

$$u(t,s) = \mathbb{E}^{\mathbb{Q}}\left[e^{-r(T-t)}(S_T - K)^+ \mid S_t = s\right].$$

The replication target

A self-financing portfolio of stock and cash must satisfy

$$\Pi_t = u(t, S_t), \qquad \Pi_T = (S_T - K)^+.$$

Matching the Brownian exposure determines the delta

Let $\Pi_t = \Delta_t S_t + \beta_t B_t$. The self-financing portfolio has diffusion term

$$\Delta_t \sigma S_t \, \mathrm{d}W_t.$$

Itô's formula gives the option diffusion term

$$\sigma S_t u_s(t, S_t) \, \mathrm{d}W_t.$$

Matching them yields

$$\boxed{\Delta_t = u_s(t, S_t).}$$

For a Black–Scholes call,

$$\boxed{\Delta_t^{\mathrm{BS}} = \Phi(d_1(t, S_t)).}$$

NO-ARBITRAGE The PDE enforces the drift; delta neutralizes the random stock exposure. Pricing and hedging are two consequences of the same model.

A delta hedge must be rebalanced

For a call, delta moves between 0 and 1:

$$\Delta^{\mathrm{BS}} \approx 0 \quad \text{deep OTM}, \qquad \Delta^{\mathrm{BS}} \approx 1 \quad \text{deep ITM}.$$

YOUR TURN A call has $S_0 = 100$, $K = 100$, $r = 0$, $\sigma = 20\%$, and $T = 1$. At inception, $d_1 = 0.10$, so $\Delta_0 = \Phi(0.10) \approx 0.54$. What position neutralizes the stock exposure of a short call?

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Answer

Hold approximately 0.54 shares per short call, and finance the residual position with cash. Rebalance after the stock, time, or model inputs change.

The same call has a different delta under local volatility

Under local volatility,

$$\frac{\mathrm{d}S_t}{S_t} = r\,\mathrm{d}t + \sigma_{\mathrm{loc}}(t, S_t)\,\mathrm{d}W_t^{\mathbb{Q}}.$$

The price still solves a one-factor PDE, but with

$$\tfrac{1}{2}\sigma^2 s^2 u_{ss} \quad \longrightarrow \quad \tfrac{1}{2}\sigma_{\mathrm{loc}}(t,s)^2 s^2 u_{ss}.$$

The hedge remains formally

$$\boxed{\Delta_t^{\mathrm{LV}} = u_s^{\mathrm{LV}}(t, S_t),}$$

but generally

$$\Delta_t^{\mathrm{LV}} \neq \Delta_t^{\mathrm{BS}}.$$

WATCH OUT The payoff did not change. The price function changed, so the hedge changed. A delta is never model-free.

A market quote does not eliminate the need for a model

A liquid vanilla call already has an observable market price. Why use a local volatility model for it?

What the market quote gives

Today's value of the option.

What the model gives

A dynamic law, a price surface between quotes, and a hedge consistent with the observed smile.

NO-ARBITRAGE Local volatility calibrates the current vanilla surface; its delta reflects the smile information ignored by a flat-volatility Black–Scholes delta.

Hedging P&L is the test performed outside the model

Suppose an option is sold and hedged dynamically. At maturity,

$$\boxed{\mathrm{P\&L} = \Pi_T - (S_T - K)^+.}$$

Source of nonzero P&L	Why it remains
Discrete trading	Rebalancing is not continuous
Numerics and costs	Greeks are approximated; trading is not frictionless
Model mismatch	The market does not follow the hedging model

YOUR TURN Does a nonzero terminal P&L prove that the delta formula was wrong?

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Answer

No. Discrete trading and model mismatch create P&L even with a correct model delta.

Model risk becomes visible under stochastic volatility

Imagine the market follows a Heston-type model,

$$\frac{\mathrm{d}S_t}{S_t} = r\,\mathrm{d}t + \sqrt{V_t}\,\mathrm{d}W_t^{(1)}, \qquad \mathrm{d}V_t = \kappa(\theta - V_t)\,\mathrm{d}t + \eta\sqrt{V_t}\,\mathrm{d}W_t^{(2)}.$$

A Black–Scholes hedge ignores both smile and random variance. A local-volatility hedge incorporates today's smile, but still makes variance a deterministic function of time and spot.

What a simulation experiment compares

Simulate the same market paths, rebalance each hedge, then compare the distributions of their terminal P&L: mean, dispersion, and tail losses.

NO-ARBITRAGE A more market-consistent model may improve hedging, but no model dominates for every contract, parameter set, or rebalancing frequency.

A lookback option: the path becomes a state variable

A lookback call remembers the highest stock level

A fixed-strike lookback call pays

$$(M_T - K)^+, \qquad M_T = \max_{0 \le u \le T} S_u.$$

Vanilla call

$$(S_T - K)^+.$$

Only the terminal stock price matters.

Lookback call

$$(M_T - K)^+.$$

The whole price path matters through its running maximum.

Same terminal stock, different path, different payoff

Take $K = 100$. Two paths both finish at $S_T = 100$:

Path	M_T	Lookback payoff
A	120	$(120 - 100)^+ = 20$
B	105	$(105 - 100)^+ = 5$

YOUR TURN Which piece of information is missing if we know only $S_T = 100$?

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Answer

The running maximum. The current or terminal spot alone cannot summarize a path-dependent payoff.

State augmentation makes the path-dependent problem Markov

At time t, define

$$M_t = \max_{0 \le u \le t} S_u.$$

The correct state is

$$\boxed{(t, S_t, M_t), \qquad S_t \le M_t.}$$

The pricing function becomes

$$\boxed{u(t, s, m) = \mathbb{E}^{\mathbb{Q}}\Big[e^{-r(T-t)}(M_T - K)^+ \mid S_t = s,\ M_t = m\Big].}$$

NO-ARBITRAGE The full past is not stored. We retain only the smallest path statistic that matters for the future payoff.

The running maximum changes only at a new record

Current state	Next stock move	What happens to M?
$S_t < M_t$	small move below M_t	M stays fixed
$S_t = M_t$	stock moves down	M stays fixed
$S_t = M_t$	stock moves up	M increases with the stock

Why this complicates a PDE

The state domain is $s \leq m$, and the diagonal $s = m$ carries a special boundary condition. The extra variable is not an independent Brownian factor; it is a constrained path functional.

Path dependence makes model choice more consequential

Vanilla option

The payoff depends mainly on the terminal distribution of S_T. Liquid vanillas constrain that distribution directly.

Lookback option

The payoff depends on hitting high levels along the way. Different models can fit vanillas today yet assign different probabilities to those path events.

NO-ARBITRAGE For exotics, model risk is not merely a question of the current smile. It is a question about future path dynamics.

Pricing a lookback option requires richer numerical state

Monte Carlo

Simulate S path by path, update

$$M_{t_{n+1}} = \max\{M_{t_n}, S_{t_{n+1}}\},$$

then discount the terminal payoff.

PDE approach

Solve for $u(t, s, m)$ on the constrained domain $s \leq m$, with a special condition on the record boundary $s = m$.

WATCH OUT A monitoring grid can miss extrema between dates. For a lookback, this discretization choice directly affects the payoff and hedge.

The lookback delta depends on history as well as spot

The hedge is still a sensitivity to the tradable stock:

$$\Delta_t^{\mathrm{LB}} = u_s(t, S_t, M_t).$$

YOUR TURN Two contracts have the same t and $S_t = 105$. Contract A has $M_t = 120$, contract B has $M_t = 108$. Should their deltas generally be identical?

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Answer

No. The established maximum changes the remaining payoff opportunity. The delta must reflect both spot and the historical record M_t.

Exotics turn model error into hedging error faster

For a lookback hedge, terminal P&L reflects the usual sources of error plus path-specific ones:

Risk	Effect
Smiledynamics	*changesprobabilitiesoffuturerecords*
Volatilitydynamics	*changesthefrequencyofextrememoves*
Monitoringfrequency	*canmissashort − livedmaximum*
Deltainterpolation	*isneededontheenlargedstategrid*

NO-ARBITRAGE The model used for an exotic should be stress-tested against the scenarios that drive its path functional, not only against vanilla prices.

Stochastic rates add a hedging factor

Random rates make discounting and the state variable random

Let r_t be a stochastic short rate. Even for a payoff $g(S_T)$,

$$\boxed{u(t,s,r) = \mathbb{E}^{\mathbb{Q}}\left[e^{-\int_t^T r_u \,\mathrm{d}u} g(S_T) \mid S_t = s,\ r_t = r\right].}$$

What changed?

The state is now (t, S_t, r_t). Rates affect the discount factor, the risk-neutral drift of the stock, and potentially the correlation structure.

Stock delta alone does not span an independent rate shock

Suppose the market has two Brownian sources: one stock-related and one rate-related.

Traded position	Stock risk	Rate risk
Stock	yes	possibly partial
Cash account	no diffusion exposure	no diffusion exposure
Zero-coupon bond	rate sensitivity	yes

YOUR TURN Can stock plus cash perfectly hedge an option exposed to an independent rate factor?

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No

Cash does not add a new Brownian exposure. A rate-sensitive risky instrument, such as a bond or swap, is needed to hedge an independent rate factor.

The natural two-factor hedge is stock plus bond plus cash

Choose a zero-coupon bond $P(t, T^\star)$ and form

$$\Pi_t = \phi_t S_t + \psi_t P(t, T^\star) + \beta_t B_t.$$

If stock and bond exposures span the two Brownian directions, choose ϕ_t, ψ_t to solve

$$\begin{pmatrix} \Gamma_S^{(1)} & \Gamma_P^{(1)} \\ \Gamma_S^{(2)} & \Gamma_P^{(2)} \end{pmatrix} \begin{pmatrix} \phi_t \\ \psi_t \end{pmatrix} = \begin{pmatrix} \Gamma_u^{(1)} \\ \Gamma_u^{(2)} \end{pmatrix}.$$

NO-ARBITRAGE Exact local replication is possible when this exposure matrix has full rank. The cash account then matches the value and drift.

Stochastic rates do not automatically imply incompleteness

Complete case

Every relevant Brownian risk is spanned by traded assets, for example stock and a sufficiently sensitive bond in a two-factor setting.

Incomplete case

There are more risk factors than independent traded exposures, for example stock plus cash but an independent short-rate factor.

YOUR TURN What determines completeness: whether rates are random, or whether tradable assets span the risk factors?

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Answer

Spanning. Random rates enlarge the hedging problem, but they do not by themselves prevent exact replication.

In an incomplete market, hedging is risk management

When perfect replication is unavailable, the objective becomes:

1. hedge stock sensitivity with the stock;
2. hedge rate sensitivity with bonds, swaps, or futures;
3. monitor and limit the residual, unspanned P&L.

A useful distinction

complete market: replication determines price

incomplete market: model and risk criterion select a hedge

NO-ARBITRAGE The complete-market benchmark remains valuable: it identifies which missing instruments or risk factors create the residual error.

Takeaways

Three case studies, one common lesson

Case	New source of complexity	Consequence for hedging
Vanilla call	model choice	delta is model dependent
Lookback option	path dependence	state and delta include the running maximum
Stochastic rates	extra rate factor	stock hedge must be complemented by bonds

NO-ARBITRAGE As contracts or models add state variables, the hedge must account for the new sources of risk or accept residual P&L.

Final check: judge the hedge where it will be used

YOUR TURN A desk calibrates a model perfectly to today's vanilla smile, then hedges a one-year lookback option daily. Which additional questions should it ask before trusting a small model price error?

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Answer

Does the model generate plausible future smile and path dynamics? How large are P&L tails under alternative models and monitoring grids? Can the required Greeks be hedged with liquid instruments?

A model price is a starting point; hedging P&L is the operational test.